Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: existential witnesses and unary generators. The author's notation correction is incorporated: the displayed kernel element depends on the constant value δ, rather than on a choice of a function c. Put

$$\Gamma_{\mathrm{def}} = \Gamma \cap \mathrm{dcl}^{\mathrm{eq}}(\mathbb{S}), \qquad \Lambda_X = \{\delta \in \Gamma_{\mathrm{def}} : \exists\ \mathbb{S}\text{-definable } c : X \longrightarrow \delta^\sharp\}.$$

Existence of c is the hypothesis guaranteeing kernel membership. After fixing δ, any two such witnesses give the same element. The witness is used only to construct the RV multiplication bijection. For the earlier full-coordinate family, write

$$g_{X,\delta,U,i} = [X]_r \otimes \big([U]_m - [U + \delta e_i]_m\big), \qquad \delta \in \Lambda_X.$$

One Gamma variable suffices for the ring kernel. In fact one can use open rays. For $a \in \Gamma_{\mathrm{def}}$, set $H_a = (a, \infty) \in \Gamma[1]$, and define

$$g_{X,\delta,a} = [X]_r \otimes \big([H_a]_1 - [H_{a+\delta}]_1\big).$$

Then in the groupified tensor ring

$$\boxed{\ker \Psi = \big(g_{X,\delta,a} : X \in \mathrm{RES}[r],\ \delta \in \Lambda_X,\ a \in \Gamma_{\mathrm{def}}\big).}$$

These generators have degree $r + 1$. The endpoint a is allowed to vary; no restriction to the single fixed ray H_0 is made. All RES degrees remain present. The fifth-root example is $a = 0$, $\delta = 1/5$.

Reduction of the higher-arity generators. Expand a semilinear Gamma set, in the scissor group, into a finite signed sum of affine rational cones. This expansion can be chosen to commute with translation. Refine the direction cones into unimodular cones, so that an integral change of coordinates turns each into a product of unary rays, whole lines and singletons, keeping all affine endpoints. The translated set has the same expansion with shifted apices. The integral coordinate matrix turns the shift δe_i into coordinate shifts $n_j\delta$ with $n_j \in \mathbb{Z}$, each witnessed by c^{n_j}. Telescope the difference of the two products.

Every whole-line difference is zero. Every singleton difference already vanishes with the coefficient $[X]$, since c^n gives $X \times b^\sharp \simeq X \times (b + n\delta)^\sharp$ and these are the existing finite-Gamma balancing coefficients. A closed-ray difference is an open-ray difference plus this singleton difference. Thus every higher-arity kernel generator belongs to the displayed unary ideal. The reverse inclusion is the original multiplication bijection. No RV lift of an affine endpoint has been chosen, no torus has been cancelled, and the total grading is preserved.

Scope of the simplification. Fixing only the translated coordinate to $i = 1$ is immediate by a parameter-free coordinate permutation, even for the positive semiring. The stronger reduction to a one-variable Gamma factor above uses a *signed* cone expansion, so it proves a generating presentation of the ring ideal; it does not by itself prove a unary presentation of the positive semiring congruence. The same witness-only correction applies to the measured notation, without asserting a measured unary reduction here.

The cone argument and its parameter and degree bookkeeping are recorded in the unary-generator verification note. The polyhedral refinement and the full kernel equality remain mathematical proofs, not completed Lean exports. No HK result is used as a premise, and the approved working manuscript remains unchanged.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.

Notation 3.19. For simplicity, we often drop the constant volume form 0 from the notation. For instance, if $\boldsymbol{U}$ is an object of $\mathrm{RV}[*]$ then it may also denote the object $(\boldsymbol{U}, 0)$ of $\mu\mathrm{RV}[*]$ with the constant volume form 0.

Notation 3.20. Let $[1] \in \mathbf{K}_+\,\mathrm{RES}[1]$ be the class of the singleton $\{1\}$. The class of the singleton $\{1\}$ in $\mathbf{K}_+\,\mathrm{RES}[0]$ is the multiplicative identity of $\mathbf{K}_+\,\mathrm{RES}[*]$ and hence is simply denoted by 1. We have $[\mathrm{RV}^{\circ\circ}_\infty] = [\mathrm{RV}^{\circ\circ}] + 1$ in $\mathbf{K}_+\,\mathrm{RV}[\leq 1]$.